Stable Distance Kernels Fail the RKBS Admissibility Condition (A4)

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Status: candidate counterexample, likely valid. For every $0 < \gamma < 1$, the kernels $\exp(-\|x-y\|_p^\gamma)$ proposed in the source fail (A4): the Euclidean kernel fails already in $\mathbb{R}^2$, and the ℓ^1 kernel fails in an explicit finite dimension depending on γ.

1 Source question and exact scope

Song, Zhang, and Hickernell define an admissible kernel in [1, Definition 1]. If $x_1, \dots, x_n, t$ are pairwise distinct, $G = [K(x_i, x_j)]_{i,j=1}^n$, and $k_t = (K(t, x_j))_{j=1}^n$, condition (A4)is

$$\left\|G^{-1}k_t\right\|_{\ell^1} \leq 1. \tag{1}$$

At the end of Section 6, on PDF page 25, the source says:

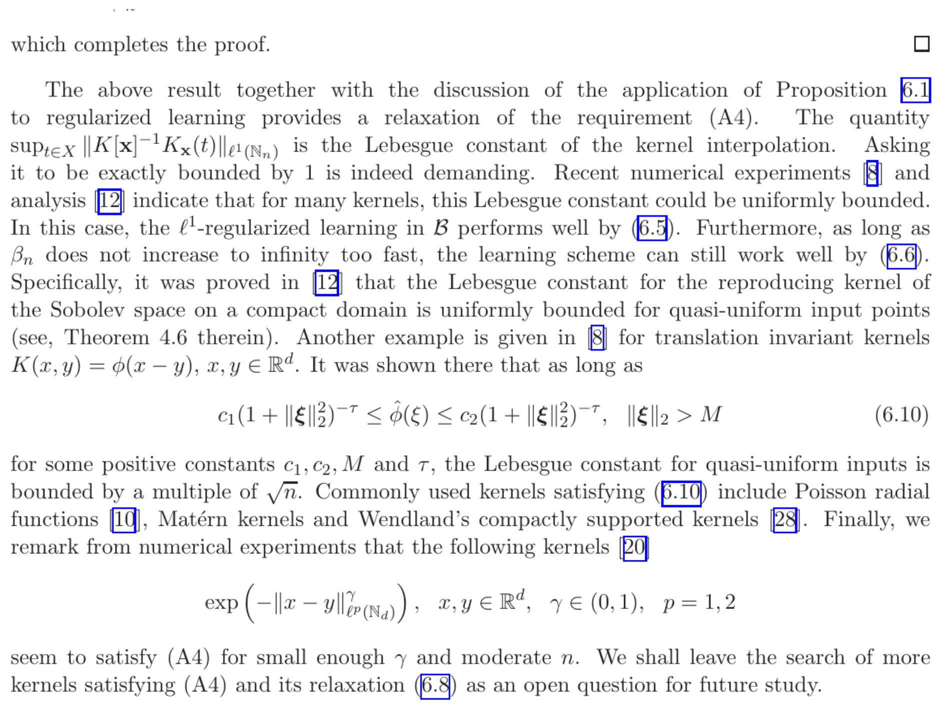
> Finally, we remark from numerical experiments that the following kernels
>
> $$\exp\Big(-\|x-y\|_{\ell^p(\mathbb{N}_d)}^\gamma\Big), \qquad x, y \in \mathbb{R}^d, \quad \gamma \in (0,1), \quad p = 1, 2,$$
>
> seem to satisfy (A4) for small enough γ and moderate n. We shall leave the search of more kernels satisfying (A4) and its relaxation as an open question for future study.

The broad request to classify more admissible or relaxed kernels is open-ended. The theorem below gives a full negative answer for the two displayed multidimensional candidate families. It does not decide their one-dimensional versions when $0 < \gamma < 1$.

2 Proof intuition

The obstacle is visible on symmetric nodes surrounding the interpolation target. Every node then has the same covariance with the target, and every row of the Gram matrix has the same sum. Thus the interpolation coefficient vector is constant, without inverting a large matrix. Its ℓ^1 norm is the quotient of two scalar exponential sums.

At zero scale, all kernel entries tend to one and this quotient tends to one. The first derivative records only the sum of powered distances among the nodes. Cross-polytopes make that derivative positive in sufficiently high dimension for both $p = 1$ and $p = 2$. In the Euclidean plane, sufficiently large regular polygons do the same for every $\gamma > 0$: their mean powered chord length is greater than the radius power. Hence the quotient crosses above one at arbitrarily small positive scales and violates (1).


which completes the proof. □

The above result together with the discussion of the application of Proposition 6.1 to regularized learning provides a relaxation of the requirement (A4). The quantity $\sup_{t\in X}\|K[\mathbf{x}]^{-1}K_{\mathbf{x}}(t)\|_{\ell^1(\mathbb{N}_n)}$ is the Lebesgue constant of the kernel interpolation. Asking it to be exactly bounded by 1 is indeed demanding. Recent numerical experiments [8] and analysis [12] indicate that for many kernels, this Lebesgue constant could be uniformly bounded. In this case, the ℓ^1-regularized learning in $\mathcal{B}$ performs well by (6.5). Furthermore, as long as β_n does not increase to infinity too fast, the learning scheme can still work well by (6.6). Specifically, it was proved in [12] that the Lebesgue constant for the reproducing kernel of the Sobolev space on a compact domain is uniformly bounded for quasi-uniform input points (see, Theorem 4.6 therein). Another example is given in [8] for translation invariant kernels $K(x,y)=\phi(x-y)$, $x,y\in\mathbb{R}^d$. It was shown there that as long as

$$c_1(1+\|\boldsymbol{\xi}\|_2^2)^{-\tau}\le\hat{\phi}(\xi)\le c_2(1+\|\boldsymbol{\xi}\|_2^2)^{-\tau},\quad \|\boldsymbol{\xi}\|_2>M \tag{6.10}$$

for some positive constants c_1,c_2,M and τ, the Lebesgue constant for quasi-uniform inputs is bounded by a multiple of $\sqrt{n}$. Commonly used kernels satisfying (6.10) include Poisson radial functions [10], Matérn kernels and Wendland's compactly supported kernels [28]. Finally, we remark from numerical experiments that the following kernels [20]

$$\exp\left(-\|x-y\|^{\gamma}_{\ell^p(\mathbb{N}_d)}\right),\quad x,y\in\mathbb{R}^d,\quad \gamma\in(0,1),\quad p=1,2$$

seem to satisfy (A4) for small enough γ and moderate n. We shall leave the search of more kernels satisfying (A4) and its relaxation (6.8) as an open question for future study.

Figure 1: The stable-distance candidates and open question on page 25 of the source paper.

3 Counterexample theorem

Theorem 1. *Fix* $0<\gamma<1$.

1. *For* $p=1$, *choose any integer* d *such that*

$$(2d-1)2^{\gamma}>2d. \tag{2}$$

   Then $K(x,y)=\exp(-\|x-y\|_1^{\gamma})$ *fails* (A4) *on* $\mathbb{R}^d$. *Consequently it fails on every* $\mathbb{R}^{d'}$ *with* $d'\ge d$.

2. *For* $p=2$, *the kernel* $K(x,y)=\exp(-\|x-y\|_2^{\gamma})$ *fails* (A4) *on* $\mathbb{R}^2$, *and hence on every* $\mathbb{R}^d$ *with* $d\ge 2$.

In both cases there are violating node sets contained in every prescribed neighborhood of the target.

The condition (2) is always achievable, since $2^{\gamma}>1$. For example, the least cross-polytope dimensions selected by (2) are $15,8,4,2$ for $\gamma=0.05,0.10,0.25,0.50$, respectively.

3.1 A symmetry lemma

Lemma 1. *Let* G *be an invertible* n*-by-*n *matrix with* $G\mathbf{1}=S\mathbf{1}$, *where* $S>0$. *If* $k_t=a\mathbf{1}$ *with* $a>0$, *then*

$$G^{-1}k_t=\frac{a}{S}\mathbf{1},\qquad \left\|G^{-1}k_t\right\|_{\ell^1}=\frac{na}{S}.$$

Proof. The displayed vector solves $Gc=k_t$. Invertibility gives uniqueness, and all its entries are positive, so its ℓ^1 norm is their sum. □

3.2 Cross-polytopes: both values of p

Proposition 1. *For each* $p \in \{1,2\}$ *and* $0 < \gamma < 1$*, the kernel* $\exp(-\|x-y\|_p^\gamma)$ *fails* (A4) *in some finite dimension. The failure occurs on arbitrarily small cross-polytopes centered at the target.*

Proof. Put $n = 2d$, take the target $t = 0$, and take the nodes

$$x_1, \ldots, x_n \in \{\pm re_1, \ldots, \pm re_d\}.$$

Write $u = r^\gamma$. In either norm every entry of k_0 is $a = e^{-u}$, and symmetry makes all Gram row sums equal.

For $p = 1$, every two distinct nodes have distance $2r$. With $b = e^{-2^\gamma u}$,

$$G = (1-b)I + b\mathbf{1}\mathbf{1}^T, \qquad S_1(u) = 1 + (n-1)e^{-2^\gamma u}.$$

The eigenvalues are $1 - b > 0$ and $S_1(u) > 0$, so G is invertible. By Lemma 1, (A4) fails exactly when

$$F_1(u) := ne^{-u} - 1 - (n-1)e^{-2^\gamma u} > 0.$$

Now $F_1(0) = 0$ and

$$F_1'(0) = (n-1)2^\gamma - n.$$

Choose $n = 2d$ satisfying (2). Then $F_1'(0) > 0$, and therefore $F_1(u) > 0$ for all sufficiently small $u > 0$.

For $p = 2$, each node has one antipode at distance $2r$ and $n - 2$ other nodes at distance $\sqrt{2}\,r$. Hence

$$S_2(u) = 1 + e^{-2^\gamma u} + (n-2)e^{-2^{\gamma/2} u}.$$

For completeness, this Gram matrix is invertible without appealing to a general kernel theorem. If P exchanges each node with its antipode and $b = e^{-2^\gamma u}$, $c = e^{-2^{\gamma/2} u}$, then

$$G = (1-c)I + (b-c)P + c\mathbf{1}\mathbf{1}^T.$$

Its eigenvalues are $S_2(u)$, $1 - b$, and $1 + b - 2c$. The last is positive:

$$\frac{1+b}{2} > \sqrt{b} = e^{-2^{\gamma-1} u} > e^{-2^{\gamma/2} u} = c,$$

where $\gamma < 2$ gives $2^{\gamma-1} < 2^{\gamma/2}$.

Again Lemma 1 reduces failure to $F_2(u) > 0$, where

$$F_2(u) := ne^{-u} - S_2(u), \qquad F_2'(0) = 2^\gamma + (n-2)2^{\gamma/2} - n.$$

Since $2^{\gamma/2} > 1$, the derivative is positive for all sufficiently large even n. Then $F_2(u) > 0$ for every sufficiently small $u > 0$. As $r = u^{1/\gamma}$ can be arbitrarily small, so can the configurations. $\square$

3.3 Regular polygons: Euclidean failure in dimension two

Proposition 2. *For every* $\gamma > 0$ *for which* $K_\gamma(x,y) = e^{-\|x-y\|_2^\gamma}$ *is strictly positive definite (in particular, every* $0 < \gamma < 1$*), the kernel fails* (A4) *on* $\mathbb{R}^2$.

Proof. For an integer $n \geq 3$, let the target be 0 and put

$$x_j = r\big(\cos(2\pi j/n), \sin(2\pi j/n)\big), \qquad j = 0, \ldots, n-1.$$

The nodes are distinct and the stable radial kernel is strictly positive definite for $0 < \gamma < 2$ by Schoenberg's theorem [2]; hence their Gram matrix is invertible.

Set $u = r^\gamma$ and

$$\delta_j = 2\sin(\pi j/n), \qquad S_n(u) = \sum_{j=0}^{n-1} e^{-u\delta_j^\gamma}.$$

The Gram matrix is circulant with row sum $S_n(u)$, while $k_0 = e^{-u}\mathbf{1}$. Lemma 1 gives

$$\left\|G^{-1}k_0\right\|_{\ell^1} = \frac{ne^{-u}}{S_n(u)}. \tag{3}$$

We first choose n. The continuous Riemann sums satisfy

$$\frac{1}{n}\sum_{j=0}^{n-1} \delta_j^\gamma \longrightarrow I_\gamma := \int_0^1 (2\sin(\pi s))^\gamma \, ds. \tag{4}$$

Moreover $I_\gamma > 1$. Indeed,

$$\int_0^1 \log(2\sin(\pi s)) \, ds = 0,$$

because $\int_0^\pi \log(\sin t) \, dt = -\pi \log 2$. The logarithm is integrable and nonconstant, so strict Jensen gives

$$I_\gamma = \int_0^1 \exp\big(\gamma \log(2\sin(\pi s))\big) \, ds > \exp\left(\gamma \int_0^1 \log(2\sin(\pi s)) \, ds\right) = 1.$$

By (4), some n therefore satisfies

$$\sum_{j=0}^{n-1} \delta_j^\gamma > n. \tag{5}$$

For this n, define $F_n(u) = ne^{-u} - S_n(u)$. Then

$$F_n(0) = 0, \qquad F_n'(0) = -n + \sum_{j=0}^{n-1} \delta_j^\gamma > 0$$

by (5). Thus $F_n(u) > 0$ for all sufficiently small $u > 0$. Equation (3) is then strictly greater than one, which violates (A4). Taking $u \downarrow 0$ places the polygon in an arbitrarily small disk. $\square$

Proof of Theorem 1. The ℓ^1 assertion follows from the first half of Proposition 1; embedding the same finite configuration by zero-padding gives failure in every higher dimension. The Euclidean assertion is Proposition 2, and the same embedding gives all $d \geq 2$. Both constructions allow r to be arbitrarily small. $\square$

4 Verification

The proof above is analytic. The deterministic script `code/verify_symmetric_counterexamples.py` independently:

- finds cross-polytope dimensions and checks the scalar quotients at 80-digit precision for six values of γ from 0.05 to 0.95;
- finds regular-polygon sizes and checks their powered-chord derivatives;
- directly builds and solves a 6-by-6 Euclidean Gram system for $\gamma = 1/2$, rather than using the quotient formula.

The direct solve reports minimum eigenvalue $7.79 \cdot 10^{-3}$, infinity-norm residual $1.11 \cdot 10^{-16}$, coefficient spread $1.69 \cdot 10^{-15}$, and (A4) norm 1.000065821370. The run ends with `ALL SYMMETRIC COUNTEREXAMPLE CHECKS PASSED`. These checks are not used as proof.

5 Limitations and bounded novelty check

The theorem settles the two named candidates negatively as multidimensional (A4) kernels. It does not classify all admissible kernels, decide either kernel on $\mathbb{R}$, or give optimal growth for the relaxed Lebesgue constants. The source's finite numerical observation for moderate n is not itself false: the theorem shows why such experiments cannot establish the universal quantifiers in (A4).

On 22 July 2026, the run's registry, solution, attempt, and proof-gap indexes were searched by arXiv id, exact title, the phrases admissible kernel, condition (A4), stable distance kernel, and the displayed exponential formula. Bounded web/arXiv searches used the exact source sentence, title/citation combinations, and the formula together with Lebesgue constant and A4. They found the source and later papers constructing other admissible RKBS kernels, including arXiv:1901.01036 and arXiv:1903.00819, but no prior resolution of this named stable-distance family. This is a bounded search, not an exhaustive novelty claim.

6 Human-review recommendation

This packet is suitable for expert review as a candidate counterexample. The highest-value checks are the derivative signs in Proposition 1, strict Jensen and the Riemann-sum quantifier in Proposition 2, and the use of strict positive definiteness for the planar Gram matrix. Until review, retain the status *candidate counterexample, likely valid.*

References

[1] G. Song, H. Zhang, and F. J. Hickernell, *Reproducing Kernel Banach Spaces with the ℓ^1 Norm*, Appl. Comput. Harmon. Anal. **34** (2013), 96–116; arXiv:1101.4388.

[2] I. J. Schoenberg, *Metric spaces and positive definite functions*, Trans. Amer. Math. Soc. **44** (1938), 522–536.